COMP233 | 4.2 Rational Numbers & Direct Proofs

DEFINITION → CLASSIFICATION → DIRECT PROOF → THEOREMS → COROLLARIES & APPLICATIONS

1 Number systems: definitions

Rational ⇔ r = a/b, a,b ∈ ℤ, b ≠0

Irrational ⇔ r ∈ ℝ and r ∉ ℚ

Every integer is rational: n = n/1.

VALID / BOUNDARY
7 = 7/1; 0 = 0/1

INVALID EXPRESSION
2/0 is undefined; neither class

2 Classification: decide carefully

Is the expression a defined real number?



NO → UNDEFINED
2/0: denominator is 0.

YES → TEST ℚ
√2 ∉ ℚ; 0.281 = 281/1000.

3 Decimal forms & special cases

Terminating decimal: 0.281 = 281/1000 ∈ ℚ

Repeating decimal: x = 0.121212...

100x = 12.121212... ⇒ 99x = 12

x = 12/99 = 4/33 ∈ ℚ

√2 is irrational (a real number not in ℚ).

4 Main theorem: closure under addition [4.2.2]

Claim: r,s ∈ ℚ ⇒ r + s ∈ ℚ

ASSUME
r,s ∈ ℚ

→

UNPACK
r=a/b, s=c/d
a,b,c,d∈ℤ; b,d≠0

→

COMPUTE
r+s = (ad+bc)/bd

→

VERIFY
ad+bc ∈ ℤ
bd ∈ ℤ and bd ≠0

Conclusion: r+s is a ratio of integers with nonzero denominator. ∴ r+s ∈ ℚ.

5 Closure operations: inspect conditions

Addition	r+s ∈ ℚ	1/3 + 1/6 = 1/2
Subtraction	r−s ∈ ℚ	5/6 − 1/3 = 1/2
Multiplication	rs ∈ ℚ	(2/3)(3/5) = 2/5
Division	r/s ∈ ℚ iff s ≠0	(2/3)/(4/5) = 5/6

COUNTEREXAMPLE: 1/0 is undefined; quotient closure needs a nonzero divisor.

6 Proof by earlier results

Theorem 4.2.1: n ∈ ℤ ⇒ n ∈ ℚ

Corollary 4.2.3: r ∈ ℚ ⇒ 2r ∈ ℚ

Reason: 2r = r+r; apply addition closure.

Derive new statements using established facts:

e.g., $3r^2-2r+4 \in \mathbb{Q}$ for r ∈ ℚ.

7 Parity: reuse known properties

Assume a is even; b is odd.

a^2 even; b^2 odd ⇒ a^2+b^2 odd

Adding 1 yields an even integer:

$a^2+b^2+1 = 2k$ for some k∈ℤ

Hence $(a^2+b^2+1)/2 = k \in \mathbb{Z}$.

8 Pitfalls & applied arithmetic

NOT a proof: test only r=1/4, s=1/2.

For universal claims, choose arbitrary r,s.

Mechanics: rates 1/6 and 1/8 job/hour.

Two hours together: 2(1/6+1/8)=7/12.

Remaining: 5/12; at 1/8 per hour:

Additional time = (5/12)/(1/8)=10/3 h.

Based on COMP233 materials and S. S. Epp, Discrete Mathematics with Applications (4th ed.), §4.2

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